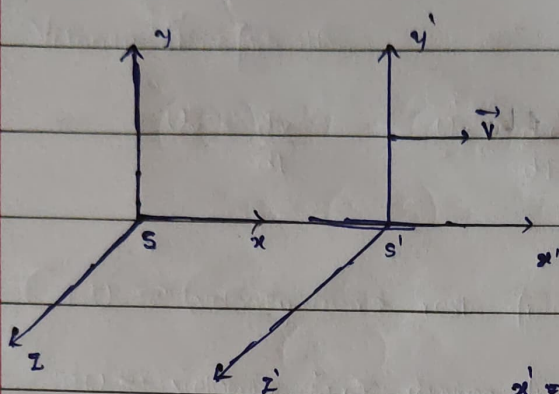


7/09/26



S' is moving with a velocity $\vec{v}$
w.r.t S along the x-direction.

$$x' = x - vt$$

$$x = x' + vt'$$

$$x' = K(x - vt)$$

$$x = K(x' + vt')$$

Objective : To find K ?

To find relation b/w t' and t ?

$$x = K(K(x - vt) + vt')$$

$$x = K(x - vt) + vt'K$$

for t', t ?

$$x = ct$$

$$x' = ct'$$

$$t' = \frac{x}{Kv} (1 - K^2 v^2) + Kt$$

(Thought exp. :- Lightning)

$$x' = ct'$$

$$K(x - vt) = ct'$$

$$\Rightarrow x = \frac{cK + Kv}{K - \frac{c}{Kv}(1 - K^2 v^2)} t$$

$$= c$$

$$\Rightarrow \tilde{\gamma}^2 = \tilde{\gamma}(\tilde{\gamma} - 1)$$

$$\Rightarrow \tilde{\gamma}^2 = \frac{c^2}{c^2 - v^2}$$

$$\Rightarrow \boxed{\tilde{\gamma} = \frac{1}{\sqrt{1 - v^2/c^2}}}$$

$$t' = \frac{x - \tilde{\gamma}^2(x - vt)}{\tilde{\gamma}v}$$

$$= \frac{x - (x - vt)(1 - v^2/c^2)}{\sqrt{1 - v^2/c^2} v}$$

$$= \frac{x - ((1 - v^2/c^2)x - (1 - v^2/c^2)vt)}{\sqrt{1 - v^2/c^2} v}$$

$$= \frac{x(1 - 1 + v^2/c^2) + (1 - v^2/c^2)vt}{\sqrt{1 - v^2/c^2} v}$$

$$= \frac{x \frac{v^2}{c^2} + vt - \frac{v^3}{c^2} t}{\sqrt{1 - v^2/c^2} v}$$

$$= \frac{x \frac{v}{c^2} + t - \frac{v^2}{c^2} t}{\sqrt{1 - v^2/c^2}}$$

$$= \frac{\frac{vx}{c^2} + t(1 - v^2/c^2)}{\sqrt{1 - v^2/c^2}}$$

$$= \frac{x - \tilde{\gamma}^2 x + \tilde{\gamma}^2 vt}{\tilde{\gamma}v}$$

$$= \frac{x(1 - \tilde{\gamma}^2) + \tilde{\gamma}^2 vt}{\tilde{\gamma}v}$$

$$= \frac{\tilde{\gamma}^2 vt - x(\tilde{\gamma}^2 - 1)}{\tilde{\gamma}v}$$

$$= \frac{\tilde{\gamma}^2 vt - x \tilde{\gamma}^2 \frac{v}{c^2}}{\tilde{\gamma}}$$

$$= \frac{t - x \frac{v}{c^2}}{\sqrt{1 - v^2/c^2}}$$

$$\therefore t' = \frac{t - x \frac{v}{c^2}}{\sqrt{1 - v^2/c^2}}$$

Velocity Transformation

$$U_x = \frac{dx}{dt}$$

$$U'_x = \frac{dx'}{dt'}$$

$U \rightarrow$ velocity w.r.t rest frame

$U' \rightarrow$ velocity w.r.t moving frame.

$$U'_x = \frac{dx'}{dt'} \bigg/ \frac{dt'}{dt}$$

$$\frac{dx'}{dt'} = \frac{U_x - v}{\sqrt{1 - \frac{v^2}{c^2}}}$$

$$\frac{dt'}{dt} = \frac{1 - \frac{v}{c^2} U_x}{\sqrt{1 - \frac{v^2}{c^2}}}$$

$$U'_x = \frac{U_x - v}{1 - \frac{v}{c^2} U_x}$$

$$U'_y = \frac{dy'}{dt'}$$

$$U_y = \frac{dy}{dt}$$

$$U'_z = \frac{dz'}{dt'}$$

$$U_z = \frac{dz}{dt}$$

$$u'_y = \frac{dy'/dt'}{dt'/dt}$$

$$u'_y = \frac{u_y \sqrt{1 - \frac{v^2}{c^2}}}{1 - \frac{v}{c} u_x}$$

$$u'_z = \frac{u_z \sqrt{1 - \frac{v^2}{c^2}}}{1 - \frac{v}{c} u_x}$$

Forward transformation ($S \rightarrow S'$) S' moves with $+v$ relative to S .

Inverse transformation ($S' \rightarrow S$) from viewpoint of S' , S moves with

velocity $-v$.

$$v \rightarrow -v$$

$$x = \gamma(x' + vt')$$

$$t = \gamma\left(t' + \frac{vx'}{c^2}\right)$$

$$u_x = \frac{u'_x + v}{1 + \frac{v}{c} u'_x}$$

$$u'_x = \frac{u_x - v}{1 - \frac{v}{c} u_x}$$

$$u_y = \frac{u'_y \sqrt{1 - \frac{v^2}{c^2}}}{1 + \frac{v}{c} u'_x}$$

$$\bar{v}_{AC} = \frac{\bar{v}_{AB} + \bar{v}_{BC}}{1 + \frac{\bar{v}_{AB} \bar{v}_{BC}}{c^2}}$$

Einstein velocity
addition rule.

Cartesian Coordinates :

$$ds^{\sim} = d\tilde{x} + d\tilde{y} + d\tilde{z} - c\tilde{dt}$$

$$ds^{\sim} = d\tilde{x} + d\tilde{y} + d\tilde{z} - c\tilde{dt}$$

(or)

$$\tilde{S} = \tilde{x} + \tilde{y} + \tilde{z} - c\tilde{t}$$

$$\tilde{S} = \tilde{x} + \tilde{y} + \tilde{z} - c\tilde{t}$$

$$= K^{\sim} (x - vt) + \tilde{y} + \tilde{z} - cK^{\sim} \left(t - \frac{vx}{c^2} \right)$$

$$= K^{\sim} \left((x - vt) - c \left(t - \frac{vx}{c^2} \right) \right) + \tilde{y} + \tilde{z}$$

$$= K^{\sim} \left(\cancel{x} + \cancel{vt} - 2\cancel{vt} - \cancel{c}t + \cancel{2vt} - \frac{\tilde{v}\tilde{x}}{c^{\sim}} \right) + \tilde{y} + \tilde{z}$$

$$= K^{\sim} \left(\tilde{x} \left(1 - \frac{\tilde{v}}{c^{\sim}} \right) + \tilde{t} (\tilde{v} - c^{\sim}) \right) + \tilde{y} + \tilde{z}$$

$$\tilde{v} - c^{\sim} = -c^{\sim} \left(1 - \frac{\tilde{v}}{c^{\sim}} \right)$$

$$= K^{\sim} \left(\tilde{x} \left(1 - \frac{\tilde{v}}{c^{\sim}} \right) - c^{\sim} \tilde{t} \left(1 - \frac{\tilde{v}}{c^{\sim}} \right) \right) + \tilde{y} + \tilde{z}$$

$$= \tilde{x} - c^{\sim} \tilde{t} + \tilde{y} + \tilde{z}$$

$$= \tilde{x} + \tilde{y} + \tilde{z} - c^{\sim} \tilde{t}$$

$$= \tilde{S}$$

∴ invariant